CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Instruction and Addressing Modes

Lecture No.- 03



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Recap of Previous Lecture



Topic

Instruction

Topics to be Covered

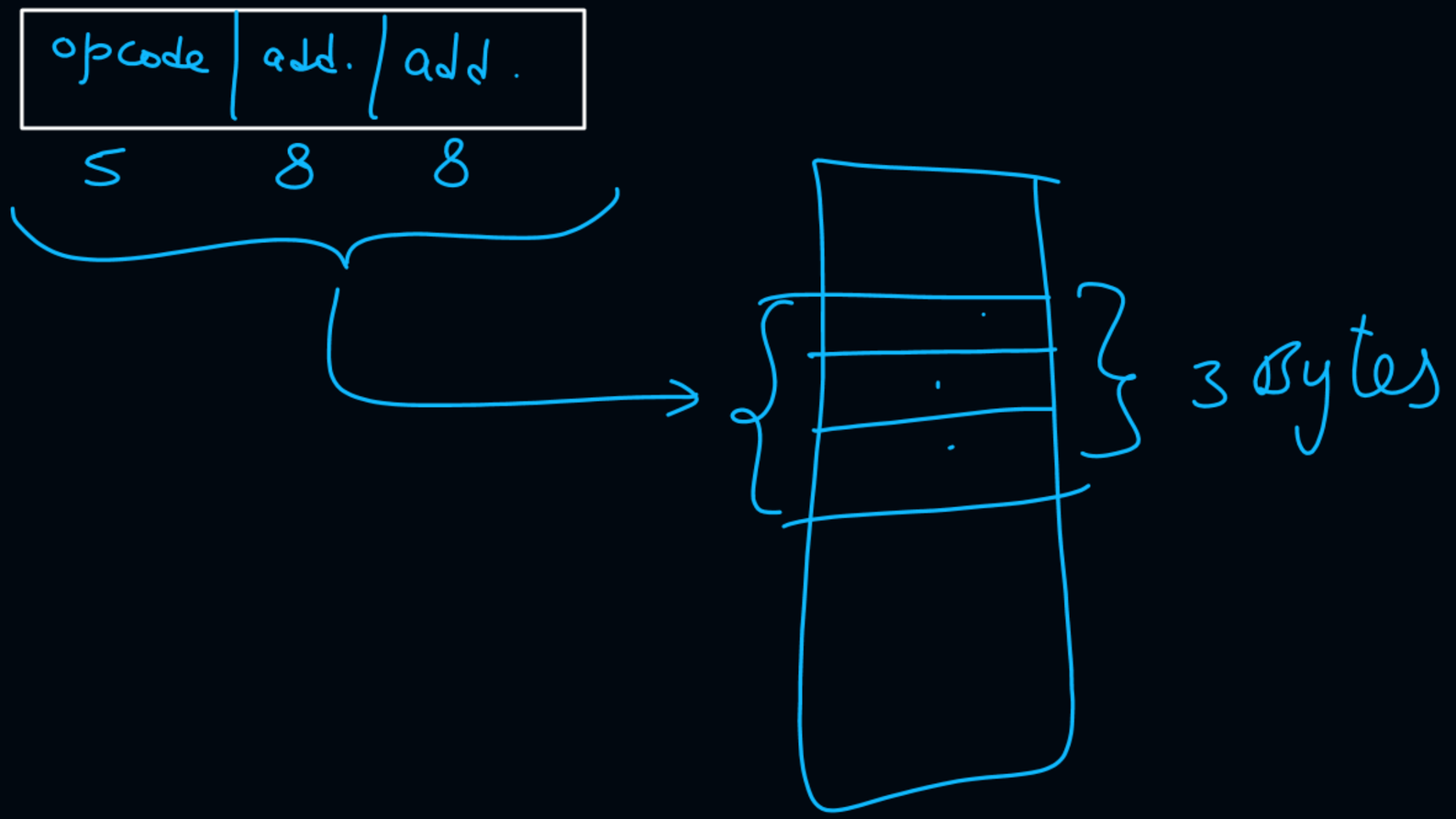


Topic Instruction

Topic Multiple Instruction Support

sign mag. and 1's comp. $-(2^{n-1}-1)$ to $+(2^{n-1}-1)$

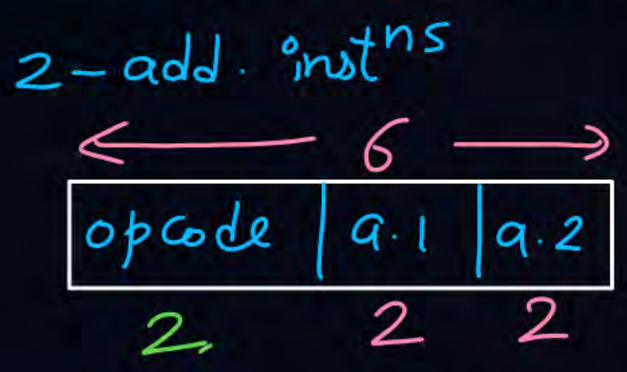
2's comp. $-(2^{n-1})$ to $+(2^{n-1}-1)$

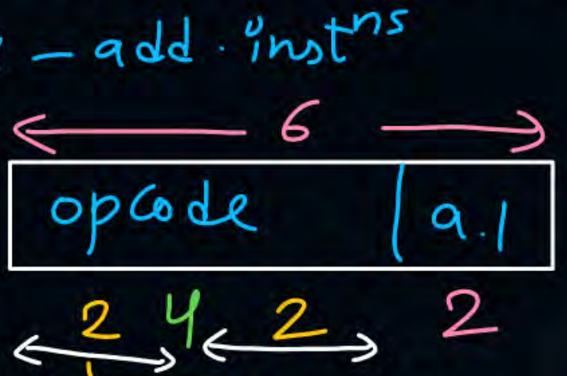

opcode
add.
add.
5
8
8
3 bytes

[NAT]

#Q. Consider a computer which supports only 2-address and 1-address instructions. Each instruction is of 6-bits and each address is of 2-bits. If there are 3 2-address instructions supported by the system then maximum number of 1-address instructions supported by system is?

Ans = 4





max opcode combinations = $2^2 = 4$ (00, 01, 10, 11)

used —11— = 3 (assume 00, 01, 10)

unused —11— = 1 (based on assumption 11)

11 00
11 01
11 10
11 11

max opcodes for 1-add. instns

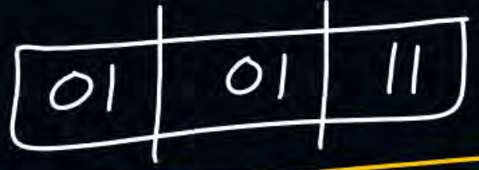
Assume CPU gets an inst^n for execution



Interpretation

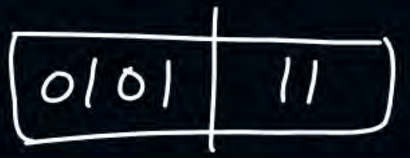
as

2 - add. inst^n



as

1 - add. inst^n



Inst^n	Interpeted as
001011	2 - add.
110011	1 - add.
100001	2 - add.
010101	2 - add.
111110	1 - add.

[NAT]

#Q. Consider a computer which supports only 2-address and 1-address instructions. Each instruction is of 6-bits and each address is of 2-bits. If there are 3 2-address instructions supported by the system then maximum number of 1-address instructions supported by system is?

In above ~~instruction~~ Question what is the range of number of 1-address instructions supported?

Ans ⇒ 1 to 4

Used 2-add. instns	unused opcodes	max 1-add. instns
4	0	$0 * 2^2 = 0$
3	1	$1 * 2^2 = 4$
2	2	$2 * 2^2 = 8$
1	3	$3 * 2^2 = 12$
0	4	$4 * 2^2 = 16$

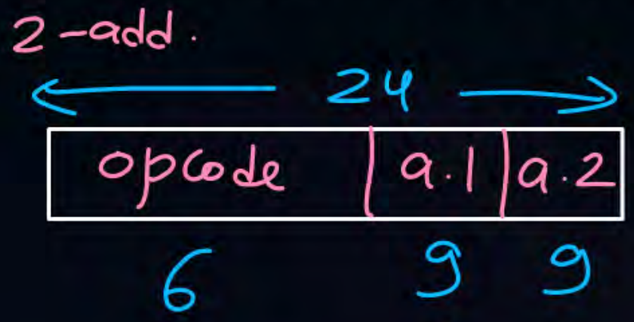
only 2-add. type instns supported

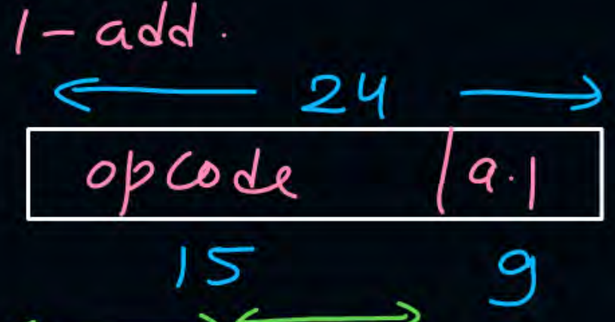
only 1-add. type instructions supported

[NAT]

Ans = 3584

#Q. Consider a system with 24-bit instructions and 9-bit addresses. If there are 57 2-address instructions then maximum how many 1-address instructions can be formulated in the system?





max opcodes = $2^6 = 64$

used opcodes = 57

unused = 7

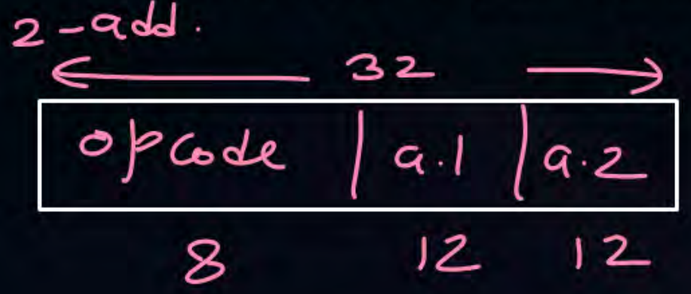
6 ⇓ 9

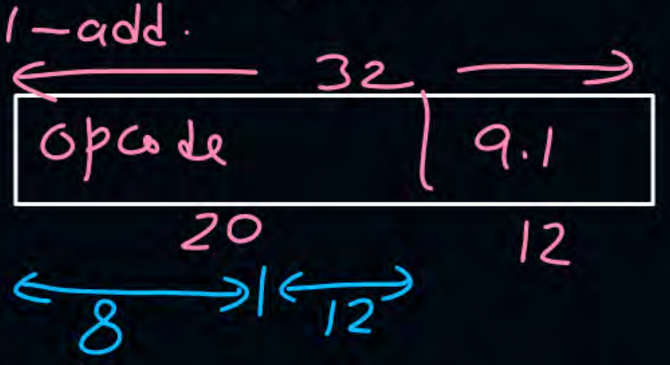
$7 * 2^9 = 3584$

[NAT]

Ans = 8192

#Q. Consider a system with 32-bit instructions and 12-bit addresses. If there are 254 2-address instructions then maximum how many 1-address instructions can be formulated in the system?





max = 2^8 = 256

used = 254

unused = 2

$2 * 2^{12} = 2^{13} = 8192$

[NAT]

Ans = $192 * 2^{12}$

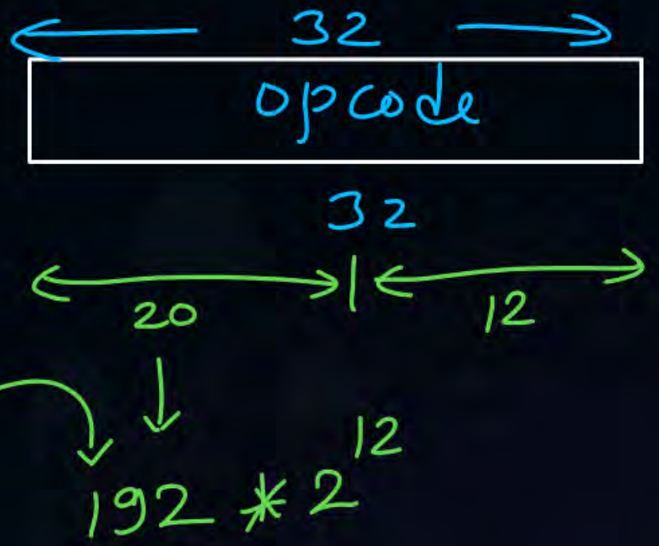
#Q. Consider a system with 32-bit instructions and 12-bit addresses. If there are 254 2-address instructions and 8000 1-address instructions then maximum how many 0-address instructions can be formulated?

from prev. question

max 1 add. instns = 8192

used = 8000

unused = 192



Ques) 3-add. , 2-add. supported

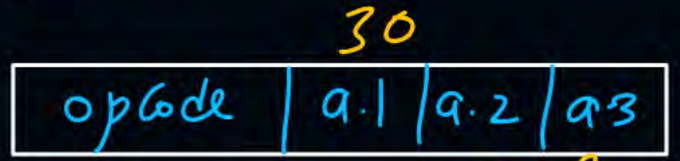
inst^n length = 30 bits

address ⇒ 8 bits

3-add. type ⇒ 32 inst^{ns}

max 2-add. inst^{ns} = 8192 ?

Soln 3-add.

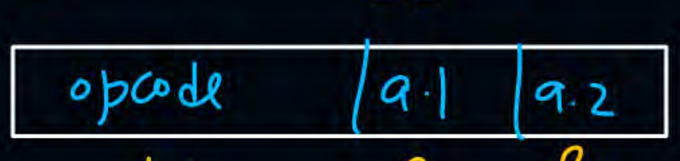


$max = 2^6 = 64$

used = 32

unused = 32

2-add.

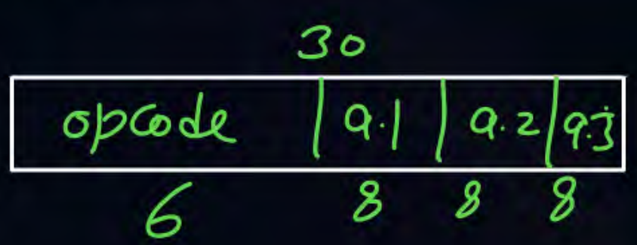


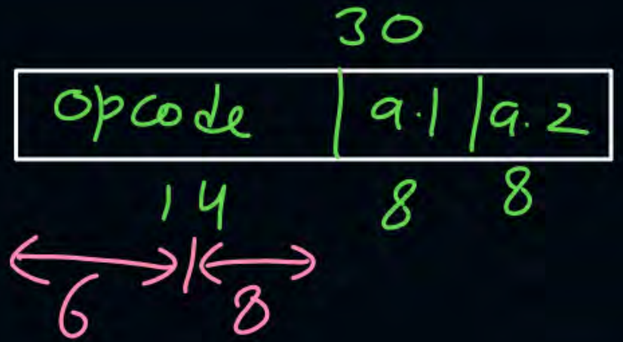
$32 * 2^8 = 2^{13} = 8192$

[NAT]

$$Ans = (64 - x) * 2^8$$

#Q. Consider a system which supports 3-address and 2-address instructions both. It has 30-bit instructions with 8-bit addresses. If there are 'x' 3-address instructions then maximum how many 2-address instructions can be formulated?





max = $2^6 = 64$

used = x

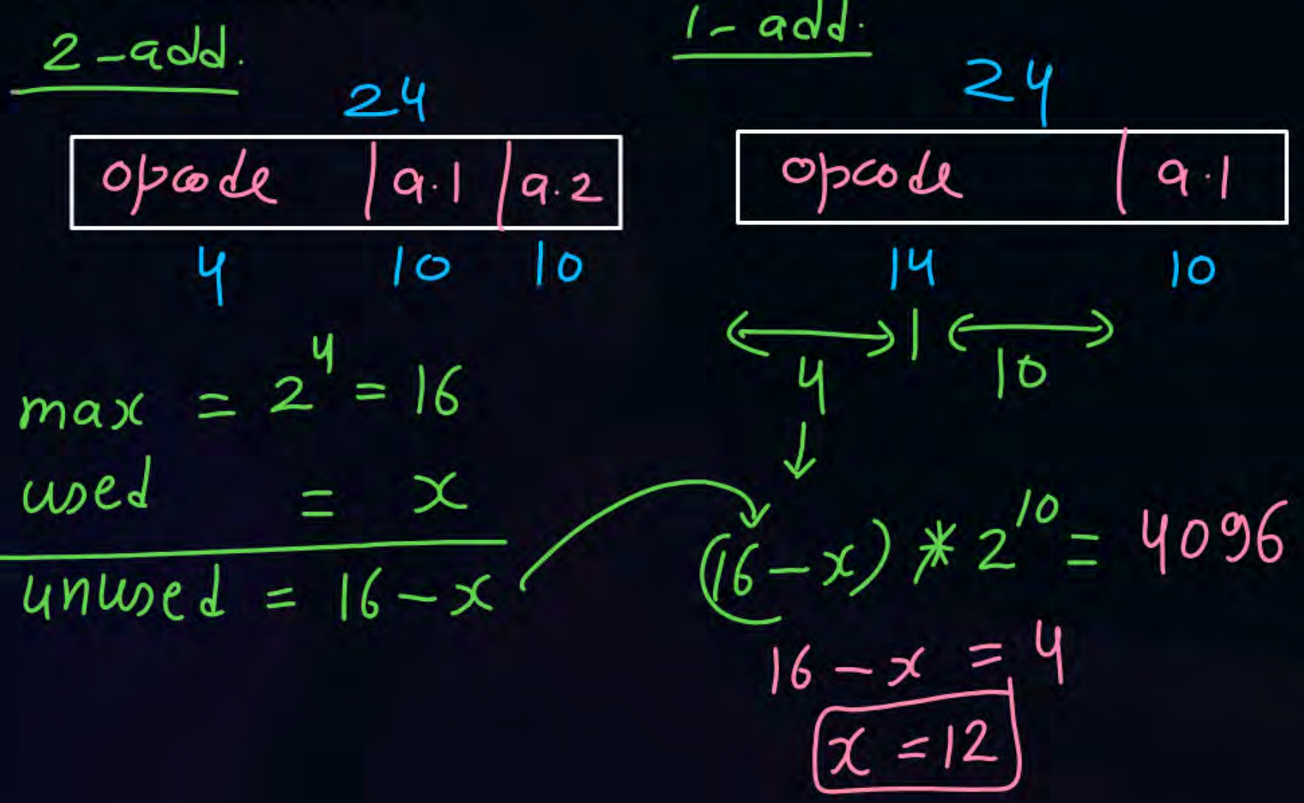
unused = $64 - x$

$(64 - x) * 2^8$

[NAT]

Ans = 12

#Q. Consider a system which supports 2-address and 1-address instructions both. It has 24-bit instructions with 10-bit addresses. If there are 4096 1-address instructions then maximum how many 2-address instructions can be formulated?



[NAT]

Ans = 48

Reg = 6 bits

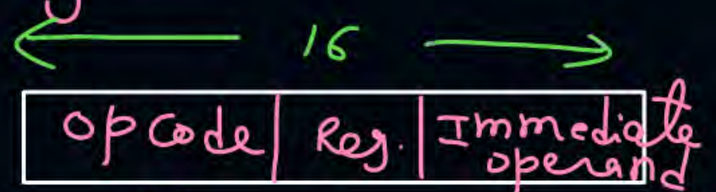
#Q. Consider a system with 16-bits instructions and 64 CPU registers. The System supported 2 types of instructions: Type-A and Type-B.

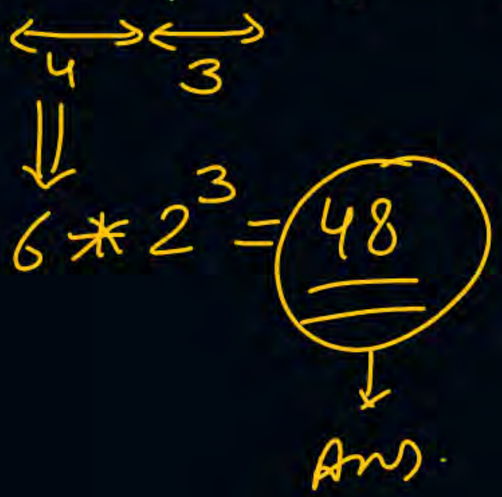
Type-A instructions have an opcode, one register operand and one immediate operand of 3-bits

Type-B instructions have an opcode, and 2 register operands.

If there are 10 Type-B instructions supported by the system then maximum how many Type-A Instructions supported by the system?

Type -A





Type - B



$$max = 2^4 = 16$$

$$used = 10$$

$$unused = 6$$

Ques) inst^n = 16 bits

3-types of inst^{ns} supported :-

1) Type 1 :- opcode, 1 Reg. operand

2) Type 2 :- opcode, 2 Reg. operands

3) Type 3 :- opcode, 1 Reg. operand, 1 Imm. operand

No. of Registers ⇒ 64

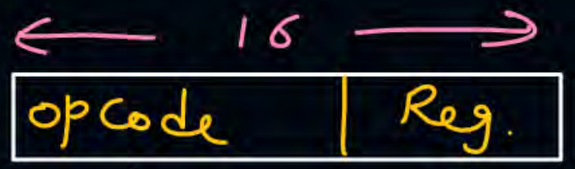
Imm. operand ⇒ 6 bits

no. of inst^{ns} ⇒ Type 2 ⇒ 4

Type 3 ⇒ 6

Type 1 = 384 ?

Type 1

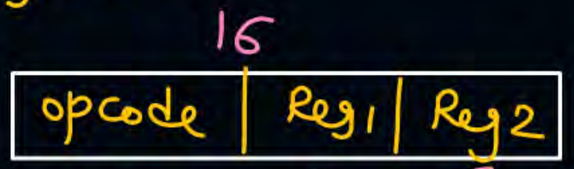




$max = 6 * 2^6 = 384$

Ans.

Type 2

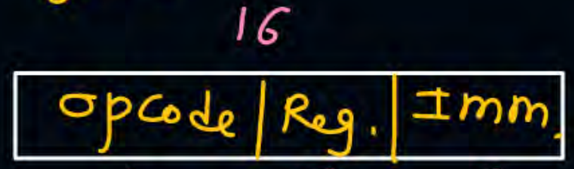


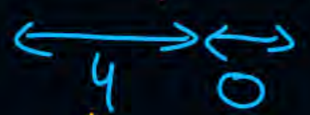
$max = 2^4 = 16$

used = 4

unused = 12

Type 3

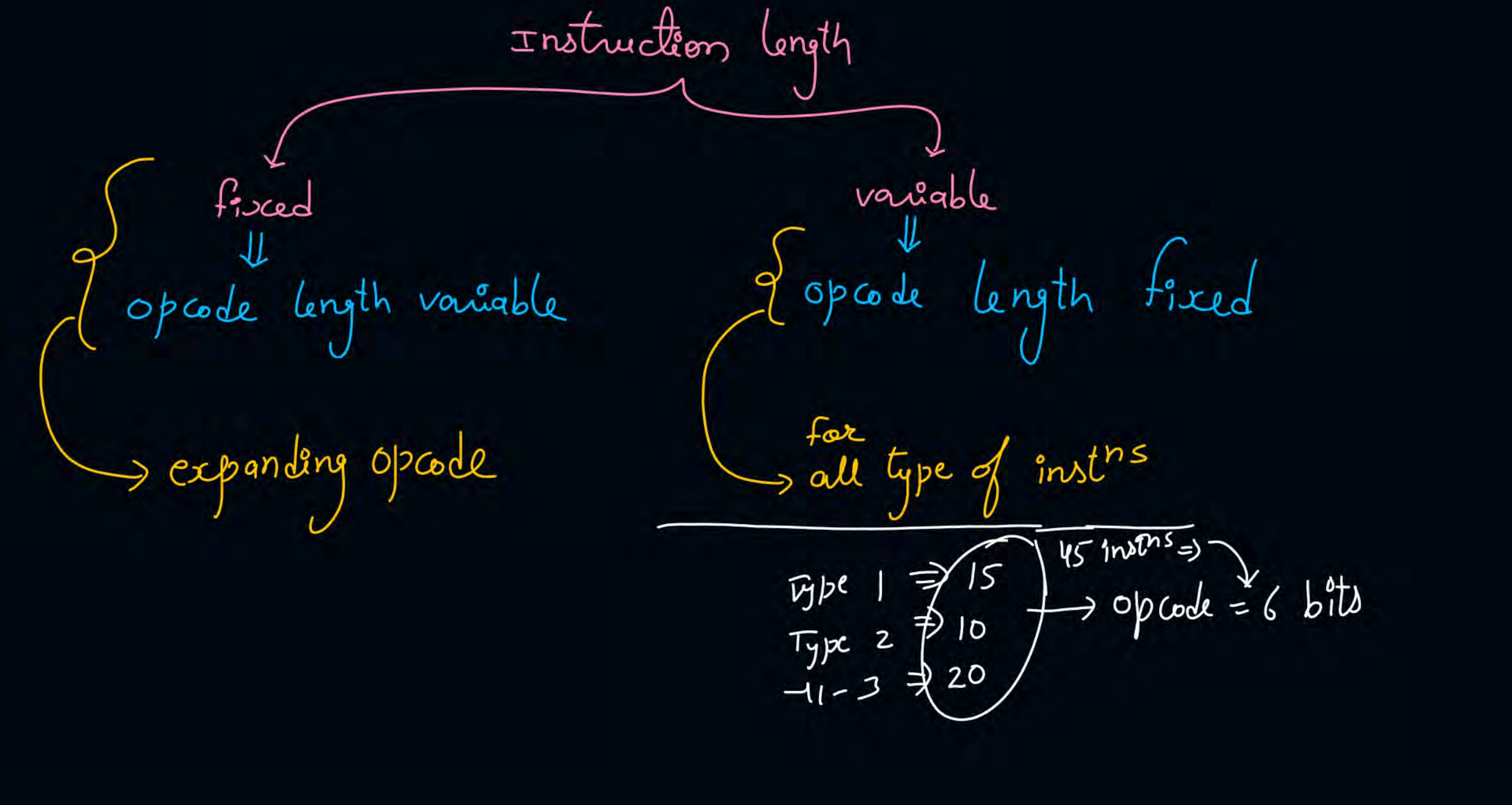




$max = 12 * 2^0 = 12$

used = 6

unused = 6


Instruction length
fixed
opcode length variable
expanding opcode
variable
opcode length fixed
for all type of instns
Type 1 ⇒ 15
Type 2 ⇒ 10
Type 3 ⇒ 20
45 instns ⇒
opcode = 6 bits

[NAT]

#Q. Consider there are 3 types of instructions in system:

1. Register Operand instructions: One opcode and 2 registers
2. Memory Operand instructions: One opcode, 1 register and 1 memory address
3. Immediate Operand Instructions: One opcode, 1 register and 1 immediate operand

Number of registers = 64 ⇒ Reg. = 6 bits

Number of bits in immediate operand = 10-bits

Memory size = 512Mbytes (byte addressable) add. = 29 bits

Total Instructions:

1. Reg Operand type: 10
2. Memory Operand type : 12
3. immediate Operand type : 4

→ Total = 26 ⇒ opCode = 5 bits

Maximum and Minimum instruction length are? 40, 17 bits Ans.

Reg. Op.



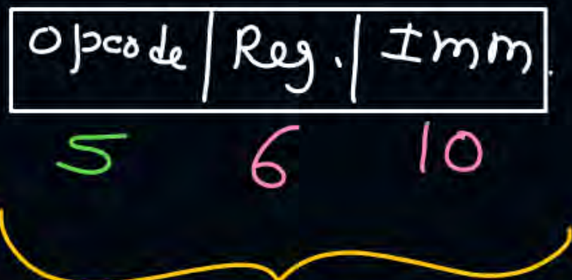
17 bits

Mem. Operand



40 bits

Imm. operand



21 bits



2 mins Summary

Topic Instructions

Topic Multiple Instruction Support



Happy Learning

THANK - YOU